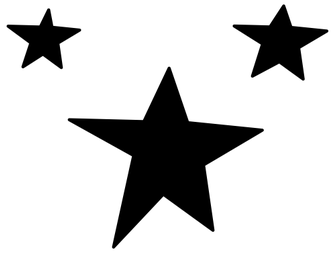
Pattern Recognition
Midterm Project (G.9)

Members:

Trinnaya

Ana

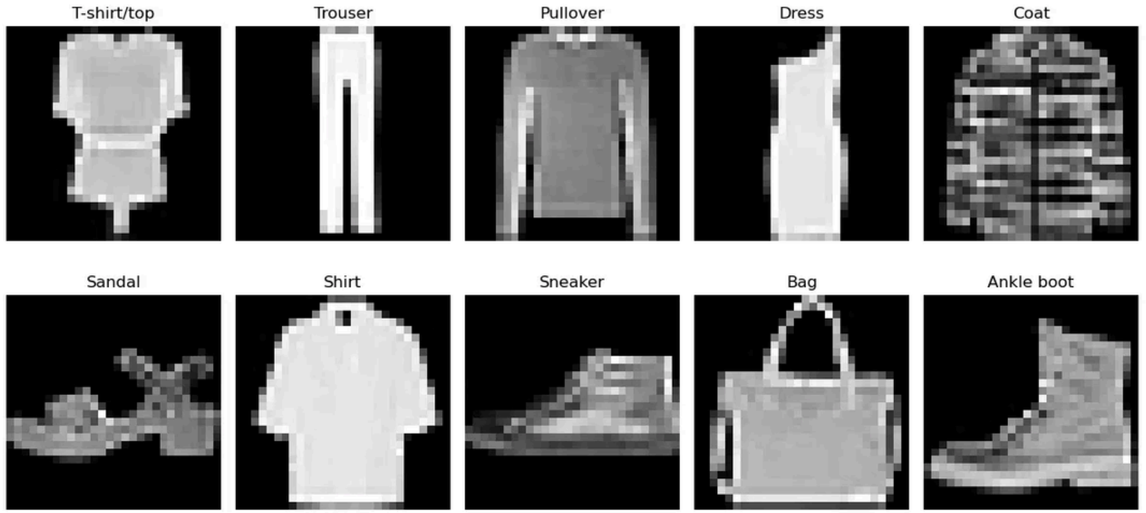
Uraiphat



The dataset

Fashion-MNIST

- 28 x 28 grayscale with pixel values 0-255
- 8,000 training, 2,000 for testing, total: 10,000 images (1,000 per class)
- 10 classes: top, trouser, pullover, dress, coat, sandal, shirt, sneakers, bag, ankle boot
- We loaded it from OpenML (fetch _ openml)



Data Preparation



- Split data into 80% training and 20% testing
- Convert each 28 × 28 image into 784 features
- Compare StandardScaler and no scaling
- Apply PCA to reduce dimensionality
- No image resizing required

Avoid Data Leakage

- Split the data before preprocessing
- Fit scaling and PCA inside the training pipeline
- Perform cross-validation using training data only
- Reserve the test set for final evaluation

Methods

- PCA: Reduces dimensionality while preserving important variance
- Logistic Regression: Linear decision model
- Perceptron: A simple linear classifier used for comparison

why choose: simple, strong, interpretable baseline

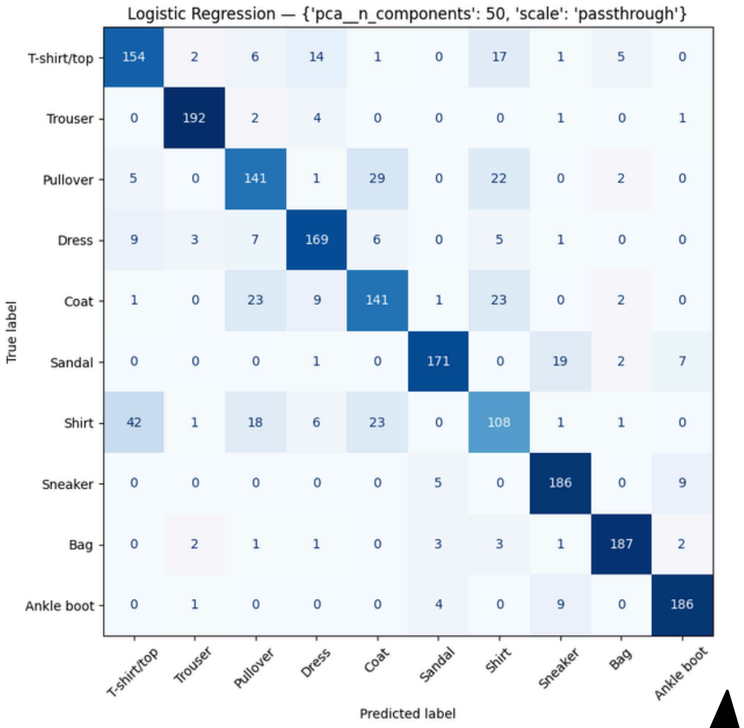


Experimental Settings

1. PCA 0.97 variance (332 components) + LogReg
2. PCA fewer components (best was 50) + LogReg
3. PCA 100 components + perceptron

Model selection

- GridSearchCV with 5-fold cross-validation
- Compare different PCA components and scaling options
- Select the best model using cross-validation accuracy



Results

Setting	CV Acc	Test Acc
PCA 100 comp + Perceptron	0.7701	0.7590
PCA 0.97 var (332 comp) + LogReg	0.8125	0.8180
PCA 50 comp + LogReg	0.8261	0.8175

Best CV Accuracy: PCA 50 + Logistic Regression

Reduced features from 784 to 50 (approximately 93.6% reduction)
Highest cross-validation accuracy: 82.61%
Test accuracy: 81.75%

Discussion

Key Findings

- PCA 50 + Logistic Regression achieved the best CV accuracy (82.61%).
- Increasing PCA components did not consistently improve accuracy.
- Shirt was the most difficult class to classify.
- Logistic Regression performed better than Perceptron.

Limitations

- Similar clothing categories are difficult to distinguish.
- PCA preserves variance, not necessarily class separation.

Future Improvements

- Try LDA for dimensionality reduction.
- Compare different scaling methods.
- Evaluate precision, recall, and F1-score.

Conclusion: PCA can significantly reduce the number of features while maintaining good classification performance.